

Sound & Waves Stations

1 Section 20.1 : The Nature of Sound

Read Section 20.1 (pp. 375-376)

1. How are sound waves produced?

2. What type of wave is a sound wave?

3. Define the following terms:
 - (a) pitch

 - (b) infrasonic

 - (c) ultrasonic

4. What sorts of materials cause sound to travel more quickly? Why?

2 Function Generator and Speaker

1. Adjust the frequency and the amplitude. Describe what you notice about the sound.

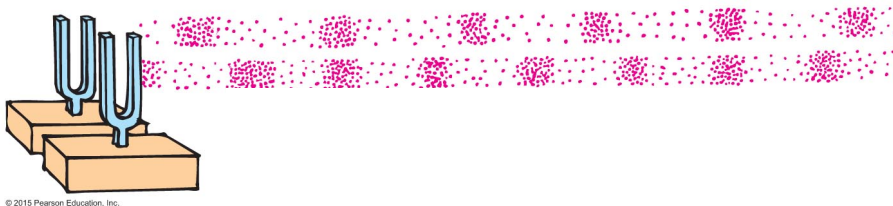
2. Start the frequency very low (2 Hz or smaller). Slowly increase the frequency until it starts to sound like a tone. This is the point at which the sound stops being *infrasonic*. Write down this frequency.

3. Start the frequency very high (Above 40,000 Hz). Slowly decrease the frequency until you can hear it. This is the point at which the sound stops being *ultrasonic*. Write down this frequency. When you are finished, **please make sure to turn the amplitude all the way down so it doesn't bother your classmates.**

3 Acoustic Beats

You will define the term “acoustic beats” in a moment. But, in the meantime, let’s experience what beats are. There are two tuning forks in front of you. One has adjustable weights. Make sure each of the weights is lined up with the indented line on the tuning fork. Hit each tuning fork. They should sound the same.

1. After having heard the two tuning forks playing the same frequency, change the frequency of the adjustable tuning fork slightly by lowering one of the weights by 1-2 centimeters. Play both tuning forks again. Explain what you are hearing.
2. This “throbbing” or “wobbling” you are hearing is called “acoustic beats.” Read Section 20.8 (pp. 385-386). What are “*beats*”?
3. In experiencing beats, why does the sound switch from loud to soft to loud to soft?
4. Move the weight even further and play the tuning forks. You should notice that the “throbbing” changes frequency. Does it shift to a higher frequency or a lower frequency?
5. To explain why beats occur, take a look at the two pendulums hanging at your station. They are different lengths, which means they have different periods and different frequencies. Set them both in motion and watch what happens. Do they ever look like they are in sync? Do they ever look like they’re out of sync?
6. Look at the diagram below (Figure 20.21 from the textbook). Label the regions of constructive and destructive interference.



4 Resonance Tube

1. Take a look at the resonance page of your notes. What is *resonance*?
2. Hit the tuning fork against the table to generate a tone. Hold it over the resonance tube. Keep changing the length of the resonance tube until you hear the sound get louder. What is happening and why?
3. Looking at your answer to #1, what do you think you are changing when you change the length of the tube?
4. Now, turn on the function generator. Gradually change the frequency. Pay attention to what happens to each of the three metal sheets. Explain why you think that they don't all resonate at the same time.

5 Slinkies

1. Send a **transverse** pulse (one at a time, and just one person) down a stretched slinky. Send a large pulse and a small pulse down the slinky. Which one appears to travel faster?
2. Send a large pulse and a small pulse down the slinky. Which one appears to travel faster?
3. Change the tension of the spring by coming closer together or moving further apart. How (if at all) does tension in the slinky seem to affect the speed of the wave?
4. Send another transverse pulse. What happens to the orientation of the wave (right side up or upside down) when it reaches the other side of the slinky and bounces back?

Name: _____

Date: _____

Period: _____

5. Stretch the large spring and send a **longitudinal** pulse down the spring. Observe the motion of the spring coils. How are they moving?

6. Have each person send a **singular wave pulse** down the spring at the same time. Make sure the amplitudes of the waves are on the **same** side of the spring.

(a) Do the waves bounce off each other or pass through each other?

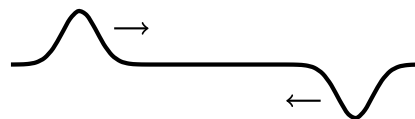


(b) What do you observe where the waves meet (in the middle)?

(c) Which kind of interference is this?

7. Have each person send a singular wave pulse down the spring at the same time. Make sure the amplitudes of the waves are on the **opposite** sides of the spring. What do you notice?

(a) Do you think the waves bounce off each other or pass right through?



(b) What do you observe where the waves meet (in the middle)?

(c) Which kind of interference is this?

8. While holding one end of the large spring firmly in place, move the other end of the spring continuously back and forth to send a continuous wave down the spring. See if you can achieve the third, fourth, and fifth harmonics. Draw your results

